

An optimisation approach for supply chain procurement transport operational planning with 3D full-truck loading constraints and returnable transport items

This paper proposes a new mixed integer linear mathematical programming (MILP) model, dubbed 3D-SCPTOP, for procurement transport planning in manufacturing supply chains. The model effectively integrates procurement and transport decisions while accounting for three-dimensional full-truck loading constraints and managing returnable transport items (RTIs). Thus, we optimise the material to procure for an assembly manufacturing site, RTIs closed-loop management, and the involved transport decisions, including the truck loading configuration. Procurement transport decisions are based on the demand of assemblers, the supplier’s inventory levels, and the dimensions of the truck / RTIs. The optimisation model aims to minimise transportation and inventory costs. As the 3D loading problem is NP-hard, a matheuristic (Math-3D-SCPTOP) approach is developed to obtain good quality solutions for large instances in an efficient computational time. The 3D-SCPTOP model was validated with a case study in an automotive seating first-tier supplier. Regarding inventory and transport efficiency, the results of the 3D-SCPTOP model are superior to the company procedures.

Keywords— Procurement, inventory, transport planning, full-truckload, 3D container loading

1 Introduction

Optimizing the use of resources in manufacturing supply chains is crucial for boosting economic growth, improving social well-being, and minimizing environmental impact. Integrating transport, inventory, and production processes is a fundamental characteristic of supply chains (Stadtler, Kilger, and Meyr 2015) because joint decisions will improve supply chain performance (Díaz-Madroño, Mula, and Peidro 2017).

Procurement refers to the process of acquiring goods and services in a supply chain (Chopra and Meindl 2015). One of the main goals of procurement is to obtain raw materials, parts or finished goods for a factory or distribution centre from a supplier. As mentioned in (Fleischmann, Kopfer, and Sürie 2015), the responsibility for transport planning usually lies with the supplier. However, sometimes the manufacturer makes transport decisions of its supplies in, for example, the automotive industry. Furthermore, an essential asset in the automotive supply chain is not considered in traditional material requirement planning (MRP): returnable containers or returnable transport items (RTI), which are used as tertiary packaging. RTIs enable the storage and transportation of RM and FG through the supply chain. Usually, the manufacturing plant supplies empty RTIs to its supplier, while the supplier delivers full RTIs with the FG or RM. Therefore, during the procurement process, transport and inventory decisions should be integrated, including RTI management. Here, we focus on optimising the operational decisions made during procurement transport planning.

According to (Díaz-Madroño, Peidro, and Mula 2014), using algorithm-based methods for procurement transport planning is more effective than relying on manual approaches based on planners’ intuition. Manual planning can be short-sighted and lead to less efficient outcomes. This becomes even more significant when optimizing procurement transport tasks that aim to reduce the number of trucks and inventory levels, especially under 3D loading constraints where the precise placement of containers within trucks must also be determined (cite). Furthermore, the capacity constraints used in the scientific literature on procurement transport planning are based mainly on freight volume or the total number of loaded product units (Hashemi-Amiri, Ghorbani, and Ji 2023) and (Díaz-Madroño, Peidro, and Mula 2014). The current research seeks to address this gap by introducing a new mixed-integer linear programming (MILP) model. This model uniquely combines procurement transportation with considerations for 3D loading, inventory management,

and RTI management. Hereafter, we refer to this model formulation as 3D supply chain procurement transport operational planning (3D-SCPTOP). Additionally, a matheuristic procedure (MATH-3D-SCPTOP) is researched.

In this paper, we present three main contributions: (i) formulating a novel MILP model, 3D-SCPTOP, that integrates procurement transport, RTI management and inventory decisions with 3D full-truck loading constraints; (ii) developing a matheuristic, Math-3D-SCPTOP, to solve the 3D-SCPTOP model in a reasonable time, which is to be used for real-world instances; (iii) validating and testing the 3D-SCPTOP model with a case study at a first-tier supplier of the automotive sector.

Readers are referred to the typology by (Wäscher, Haußner, and Schumann 2007) for a global view of container loading problems, where the different dimensions of the loading problem are described. Furthermore, (Silva, Toffolo, and Wauters 2019) summarise the exact methods used in the literature to solve container loading problems, and (X. Zhao et al. 2016) explore other solution approaches in the literature. Readers are also referred to the literature reviews of (Mosca, Vidyarthi, and Satir 2019) and (Díaz-Madroño, Peidro, and Mula 2015), which focus more on integrating inventory decisions and transport. Here, we jointly address container loading, procurement transport and inventory decisions. [Furthermore, we refer readers \(\) for insights about the packaging methods used in supply chains.](#)

The paper is organised as follows. Section 2 provides a literature review related to the problem under study. Section 3 presents the problem description. Section 4 describes the 3D-SCPTOP MILP formulation and the developed matheuristic solution approach, MATH-3D-SCPTOP. Section 5 validates and experiments with the proposal in a real case study. Finally, Section 6 concludes and puts forward further research lines.

2 Literature Review

The goal of this research is to integrate procurement transport operational planning decisions with 3D loading restrictions and RTI packaging. Based on the authors' research, no previous work has been identified that meets this criterion. Therefore, three independent, but sometimes overlapping, knowledge bodies have been reviewed: procurement transport and inventory planning, container loading problems and RTI literature.

2.1 Procurement transport and inventory planning

Models that integrate transport and inventory costs during the procurement process are widely studied in the literature. As noted in (Langley et al. 2020), transportation facilitates the movement of raw materials and finished products between suppliers, warehouses, manufacturers, and retailers. Therefore, transport processes play a crucial role in supply chains. (Fleischmann, Kopfer, and Sürie 2015) distinguish between two main types of transportation within a supply chain: "procurement transport," which involves moving materials from suppliers to manufacturing sites, and the "distribution of goods," which concerns delivering finished products from factories to customers. This article focuses specifically on addressing the challenges related to procurement transportation.

There are two main dimensions of procurement transport problems according to the supply chain configuration and the applied transport method. Regarding supply chain configurations (Mosca, Vidyarthi, and Satir 2019): (i) single source to a single destination (SS); (ii) multiple sources to a single destination (MS); (iii) single source to multiple destinations (SM); (iv) multiple sources to multiple destinations (MM). Source components determine the number of supplier locations that are taken into account, while destination components denote the number of supplied manufacturers. Additionally, different transport methods are identified in (Díaz-Madroño, Peidro, and Mula 2014): milk run, less-than-truckload (LTL), full-truckload (FTL). We also analyze the formulation of the transportation capacity constraints in the scientific literature.

(Berman and Wang 2006) address the problem of optimising distribution strategies to deliver products from suppliers to plants, whose aim is to minimise transport, pipeline and plant inventory costs by using cross docks. A mixed integer non-linear programming (MINLP) formulation, a Lagrangian relaxation heuristic, a greedy heuristic, and a branch-and-bound algorithm are proposed to model and solve the problem.

(Sancak and Salman 2011) present a MILP model to optimise procurement transport decisions in a SS supply chain configuration using LTL and FTL transport methods. The objective is to minimise transport and inventory holding costs, while fulfilling production plan requirements on a multiperiod planning horizon. The study explores the possibility of delaying LTL shipments to the subsequent time period, and utilising safety stocks whenever necessary, in a stochastic environment by employing data from a bus manufacturer. The findings demonstrate that delaying shipments leads to reduced holding and transport costs with a minimal stockout risk.

(Schöneberg, Koberstein, and Suhl 2010) examine the concept of delivery profiles, which define the delivery frequency at a fixed rate, offering structured planning, minimizing management effort, and improving coordination in logistics. In this context, FTL transport is used when the goods' volume exceeds a vehicle's capacity, making direct shipment possible. LTL is applied when the volume is insufficient to fill a truck, and the goods are first sent to a consolidation center before reaching the final destination. In (Schöneberg, Koberstein, and Suhl 2013), the proposed model accounts for demand uncertainty.

(Díaz-Madroño, Mula, and Peidro 2014) present a fuzzy multi-objective integer linear programming (FMOILP) model that focuses on operational procurement transport planning in an automotive supply chain, which involves a single source and a single destination while utilising FTL. The developed model aims to minimise the number of trucks employed for procurement transport and the inventory level by considering the uncertainties related to transport capacity levels.

(Hein and Almeder 2016) investigate the benefits of integrated procurement transport and production planning for supply and production processes by comparing a sequential approach, where production is firstly determined, followed by procurement transport decisions, plus an integrated approach in which decisions are jointly made. Numerical experiments reveal that the integrated planning approach leads to substantial cost savings.

(Díaz-Madroño, Mula, and Peidro 2017) present a novel MILP model that integrates procurement, production and transport planning decisions within manufacturing systems. This model extends beyond traditional material requirement planning (MRP) systems by incorporating procurement transport decisions from multiple sources and a single destination into a unified optimization framework. It considers the capacities of both material and production resources, along with procurement transport planning decisions, by employing various shipping modes such as FTL, LTL, and milk-run.

(Chitsaz, Cordeau, and Jans 2019) address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces a novel MILP model for the ARP and proposes a three-phase decomposition matheuristic for its solution.

(Baller et al. 2022) examine the optimization of procurement transport in an automotive factory, focusing on first-tier suppliers at an aggregate level across multiple sources (570) to a single destination, using FTL, LTL, and courier express service (CES) as transportation methods. The proposed MILP approach includes a cyclic inventory constraint, truck utilization, and storage limitations.

(Hashemi-Amiri, Ghorbani, and Ji 2023) propose an integrated approach to optimize a perishable food supply chain. The model combines supplier selection, procurement and distribution transport, production scheduling, and vehicle routing to improve supply chain efficiency and resilience. It handles demand uncertainties through a distributionally robust chance-constrained model. The model is reformulated as MILP by utilizing duality and linearization techniques.

2.2 Procurement Transport with RTIs

Some models further integrate the procurement and delivery of RTIs into the procurement transport literature. In this sense, RTIs are procured to the supplier, who loads the items and sends them to customer. We extend the SC definitions previously defined to contemplate the RTIs management strategies.

model a food closed-loop supply chain network design problem incorporating returnable transport items (RTIs), product perishability, facility location and demand uncertainty. A two-stage non-linear chance-constrained programming model minimizes fixed facility, production, transport, inventory holding and RTI costs while ensuring a target service level. It's linearized via sample average approximation and distribution-free methods to capture forward RTI-based shipments and reverse flows for reuse. To solve large instances, an improved Lagrangian relaxation algorithm decomposes facility-location and flow decisions, outperforming CPLEX on solution quality and speed.

(Soysal et al. 2023) develop a tactical-level, multi-period closed-loop Vendor-Managed Inventory (VMI) system for RTIs in an agro-food SC. The SM SC integrates forward flows of full RTIs from a single vendor to multiple customers and reverse flows of empties back to the vendor for cleaning . All routing and inventory decisions are made centrally by the vendor under full information sharing. Sustainable plans in terms of amounts of energy spent and empty crates collected are devised and evaluated.

(Guo et al. 2024) model two-tier Container Management Centre (CMC) network . Empty RTIs are dispatched from central CMC hubs to upstream suppliers, loaded with products, carried by truck to downstream customers, then returned empty to regional CMCs for cleaning and storage before being shipped back to central hubs. A neural network is used for demand forecasting. A MILP is proposed that maximises total profit by optimally determining transport costs and the penalties applied to late RTIs returns.

(Y. Zhang et al. 2024) formulate a perishable-food inventory-routing MILP with multiple RTI types in a SM supply chain, integrating procurement of products and pickup of both filled and empty RTIs. Decisions

are made on a operational level, optimally determining per period the quantities to pack, deliver, pick up, purchase and lose, as well as the vehicle routes in order to maximise the total profit of the SC that is equal to the total quality-dependent revenue minus the total costs involved, including inventory holding costs, product cost, transportation costs, and RTI purchasing and loss costs. The model is solved exactly via CPLEX solver and approximately via a tailored Kernel Search matheuristic.

2.3 Container loading

Container loading problems encompass a subset of geometric assignment problems known as cutting and packing ($C\&P$) problems. They revolve around the allocation of n -dimensional orthotope items in larger n -dimensional orthotope containers. Here the aim is to optimise an objective function while taking into account two key geometric constraints: complete containment of items inside containers and lack of item overlap. A loading pattern serves as a formal representation of a solution to such an assignment problem. Dyckhoff introduced in (Dyckhoff 1990) a typology that unifies $C\&P$ problems using four different criteria: dimensionality of space, the objective of assignment, containers types and box type. Consequently, (Wäscher, Haußner, and Schumann 2007) develop a new typology that has become the standard in the $C\&P$ literature. We briefly describe the typology used in (Wäscher, Haußner, and Schumann 2007) based on five different criteria: (i) dimensionality of space; (ii) kind of assignment (input minimisation of containers for all small items and loading patterns; output maximisation to pack a subset of small items with limited container availability;); (iii) assortment of small item by distinguishing if they are either a weakly (or strongly) heterogeneous assortment or identical small items ; (iv) assortment of large objects (if it is either several large objects or one large object); (v) shape of small items (circular or of another shape). Table 1 links the standard $C\&P$ nomenclature and the criteria of the typology according to (Wäscher, Haußner, and Schumann 2007).

Table 1: $C\&P$ nomenclature and criteria

Assortment of small items		Identical	Weakly heterogeneous		Strongly heterogeneous	
Objective		Output maximization	Input minimization	Output maximization	Input minimization	Output maximization
Assortment of large items						
Fixed Dimensions	Identical	X	Single Stock Size Cutting Stock Problem (SSSCSP)	Multiple Identical Large Object Placement Problem (MILOPP)	Single Bin Size Bin Packing Problem (SBSBPP)	Multiple Identical Knapsack Problem (MIKP)
	Strongly heterogeneous		Residual Cutting Stock Problem (RCSP)	Multiple Heterogeneous Large Object Placement Problem (MHLOPP)	Residual Bin Packing Problem (RBPP)	Multiple Heterogeneous Knapsack Problem (MHKP)
	Weakly heterogeneous		Multiple Stock Size Cutting Stock Problem (MSSCSP)		Multiple Bin Size Bin Packing Problem (MBSBPP)	
	One large object	Identical Item Packing Problem (IIPP)	X	Single Large Object Placement Problem (SLOPP)	X	Single Knapsack Problem (SKP)
Variable dimension	One large object	X	Open Dimension Problem (ODP)	X	Open Dimension Problem (ODP)	X

The first mathematical programming model in a scientific journal to describe the container-loading problem was developed in (C. S. Chen, Lee, and Shen 1995), where the authors introduced an MILP model to address the 3D input minimisation problem by considering rotations and containers of varying dimensions. However, it does not specify the use of weakly or strongly heterogeneous boxes or containers. Additionally, real-world constraints are modelled like, for example, weight distribution.

(Junqueira, Morabito, and Sato Yamashita 2012b) propose an alternative MILP formulation to (C. S. Chen, Lee, and Shen 1995) using binary encoding for box position, which utilises normal patterns to reduce possible box coordinates by considering multiples of boxes dimensions and ensuring optimal solutions. Additionally, loading and cargo constraints are developed to ensure the static and dynamic stability of the cargo. Boxes must be supported by other box top faces or the container floor. The same authors integrate further loading restrictions in (Junqueira, Morabito, and Sato Yamashita 2012a), where they integrate multidrop

constraints, which address the challenge of packing a truck with boxes for different customers at various destinations. The goal is to plan the container’s loading to minimise the time spent unloading and reloading boxes at each destination.

(W. Zhu and Lim 2012) devise a tree search-based method to solve SLOPP and SKP with 3D constraints. Their approach involves evaluating free spaces by employing a look-ahead tree search and selecting optimal placements based on the best free space-block pairs.

(Gonçalves and Resende 2013) develop a biased random-key genetic algorithm (BRKGA) designed to solve SBSBPP. The algorithm addresses geometric constraints and introduces a maximal-space representation to handle available free spaces in bins. A novel placement procedure is hybridised with the BRKGA, which optimises the order of box packing and the parameters utilised by the placement procedure to offer an effective solution approach to an NP hard problem.

(Bortfeldt and Wäscher 2013) identify constraints for *C&P*, including item-related constraints (orientation constraints, load bearing, loading priorities), container-related constraints (weight distribution and weight limit), and cargo-related constraints (complete-shipment constraints) and load-related constraints (complexity constraints or stability constraints). The study also highlights the achievements and deficits in container loading research that pertain to these constraints.

(Paquay, Schyns, and Limbourg 2016) propose an MILP model for the MBSBPP. One noteworthy constraint is the assumption that containers are truncated parallelepiped instead of rectangular boxes. They also consider constraints, such as box rotation, container load bearing, weight distribution, stability and fragility of items.

(Mahvash, Awasthi, and Chauhan 2018) introduce a novel resolution procedure for the SBSBPP through the column generation technique, where the columns that are generated are the loading configurations of a single bin. For the pricing develop problem they develop a heuristic to find new columns with a negative reduced cost. This approach not only produces better results than the previous articles on the literature, but also gives new lower bounds.

(Silva, Toffolo, and Wauters 2019) study SSLOP models with 3D constraints. These authors compare the different approaches that exist in the literature for model packing problems based on two fundamental approaches: discretised formulation (Junqueira, Oliveira, et al. 2013) and continuous formulation (C. S. Chen, Lee, and Shen 1995) with subsequent modifications. They also indicate that discretised formulation (Junqueira, Morabito, and Sato Yamashita 2012a) performs the best for instances with similar sized boxes. However, it encounters memory limitations for large instances. For other sets of instances, the adapted formulations of (C. S. Chen, Lee, and Shen 1995) and (Paquay, Schyns, and Limbourg 2016) outperform other adaptations. Further research topics suggest including the design of new formulations by considering the limited approaches in the literature.

(Kurpel et al. 2020) introduce techniques and exact approaches to solve input minimisation and output maximisation problems for container-loading problems. Their methods consider practical constraints, such as orientation, separation of boxes and vertical stability. A set partitioning method is also employed in order to obtain upper bounds in a container-loading input minimisation problems. The authors adapt and evaluate four discretisation methods found in the existing literature and incorporate mathematical formulations to address the aforementioned constraints.

(Alonso, Alvarez-Valdes, and Parreño 2020) develop an algorithm by the greedy randomised adaptive search procedure (GRASP) to tackle a practical container-loading problem with time-sensitive constraints. Their approach addresses dynamic stability constraints while stacking products on pallets and loading them onto trucks during different time periods. Their approach provides high-quality solutions and surpasses an integer programming formulation in efficiency and effectiveness terms.

(Ali et al. 2022) conduct a comprehensive literature review by identifying constraints and solution methods for offline problems with full knowledge of the items to be packed and those tackling the online problems, where items are packed immediately when they arrive, and no prior knowledge of an item is known. The study also identifies and classifies solution approaches for offline packing into six categories: exact methods, constructive heuristics; local search and metaheuristic methods; approximation algorithms; machine learning and hyper-heuristic methods; tree search methods. (Harrath 2022) proposed a three stage heuristic based on layer generation to find good quality solutions the SBSPP problem for large instance up to 993 bins and 200000 items taking into account load balancing constraints.

(Tresca et al. 2022) develop a matheuristic to solve the SBSBPP that includes items with different IDs and different categories, to be delivered to different clients. The algorithm creates monoitem, monocategory, and mixed layers which then are used to compute the final loading configurations of the bins.

(Li, M. Chen, and Huo 2022) develop a hybrid adaptive large neighborhood search algorithm to solve the multiple container loading problem, which is tries pack a given set of items with the minimum number of containers while maximizing the space utilization of them. Therefore, it is both a MBSBPP and a MSSCSP.

Their approach takes into account some practical applications such as suspension constraints, rotation and weight limit.

(M. Chen, Huo, and Duan 2023) present a new approach to the 3D-BPP, introducing the 3D-BS DPP which focuses on optimizing both the design of bin sizes and the packing process. A novel Hybrid Biogeography-Based Optimization (HBBO) algorithm is proposed, integrating differential evolution and a local search strategy for enhanced performance.

One noteworthy trend in $C\&P$ is integrating loading problems with routing. This is known as capacitated vehicle-routing problem with 3D or 2D loading constraints (3L-CVRP or 2L-CVRP). (Junqueira, Morabito, and Sato Yamashita 2012b) present a MILP model with binary encodings for 3L-CVRP by including loading, cargo and multidrop constraints. The combination of these two NP-hard problems requires using heuristics and metaheuristics for real-world instances, as seen in (Dominguez et al. 2016), (Alonso, Martinez-Sykora, et al. 2022) and (Wei et al. 2015). Another noteworthy trend is to use reinforcement learning techniques to solve $C\&P$, as in (Que, Yang, and D. Zhang 2023) and (H. Zhao, C. Zhu, and Huang 2022).

2.4 Research Gaps & Methodology

Here, we discuss the main differences between the 3D-SCPTOP model proposal and those reviewed by specifically focusing on procurement transport. The key distinction of the 3D-SCPTOP model compared to the existing literature is the incorporation of 3D constraints and vertical stability constraints to model transport capacity with a view to providing more realistic solutions and reducing the likelihood of infeasible outcomes. Table 2 presents a comparison of inventory modelling, transport modelling, and supply chain configuration between the 3D-SCPTOP model proposal and those currently in the literature. Hence, the 3D-SCPTOP approach is based on a SS supply chain configuration, and takes into account the FTL transportation mode.

Article (Guo et al. 2024)	SC Configuration	Decision level	Transportation Mode	Transportation Capacity Constraint	Transportation Cost	Inventory Capacity Restriction	RTIs	Inventory Costs	Safety Stocks	Modelling Framework
	MM	TO	FTL	UH	X	X	S—NF	X		MINLP
(Berman and Wang 2006)	MM	T	FTL—LTL	V	X			X		MINLP
(Sancak and Salman 2011)		O	LTL—FTL	V	X			X	X	MILP
(Schöneberg, Koberstein, and Suhl 2010)	MS	T	FTL—LTL	V	X	X		X		MILP
(Schöneberg, Koberstein, and Suhl 2013)	MS	O	FTL—LTL	V	X	X		X		MILP
(Díaz-Madroño, Mula, and Peidro 2014)	SS	O	FTL	UH	X			X	X	FMOILP
(Hein and Almeder 2016)	MS	O	MR	V	X			X		MILP
(Díaz-Madroño, Mula, and Peidro 2017)	MS	T	FTL—LTL—MR	V	X	X		X	X	MILP
(Meyer and Amberg 2018)	MS	T—O	MR—FTL	V	X	X		X		MILP
(Chitsaz, Cordeau, and Jans 2019)	MS	O—T	MR	V	X	X		X		MILP
(Baller et al. 2022)	MS	O	MR—FTL—LTL—CES	V—W	X	X			X	MILP
(Hashemi-Amiri, Ghorbani, and Ji 2023)	MS	T	MR	V	X					CCP
3D-SCPTOP	SS	O	FTL	3D-Constraints	X	X		X	X	MILP
(Y. Zhang et al. 2024)	MS	TO	MR	UH	X	X	M—NF	X		ILP

Abbreviations. **T** (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight), **UH** Unit Holding; **CCP** chance-constrained programming

Table 2: Comparison of procurement transport models

On the other hand, the 3D-SCTOP model diverges from those proposed in the packaging literature in several aspects. Firstly, the loading problem takes into account different time periods. Therefore, the specific container (small items or boxes in the $C\&P$ literature) that we utilise are yet to be defined, as the choice of container depends on the inventory levels of each product. Also, the objective function depends on inventory and transport costs, which are inherently connected to the loading problem at hand. Therefore, we simultaneously solve an input minimisation and output maximisation problem. Both of these are a result of transport minimisation and inventory costs. Input minimisation comes because we want to find the minimum number of trucks needed. Meanwhile, output maximization comes from the FTL and the transport costs.

Based on the literature review and the identified research gaps, our proposal is structured according to the steps depicted in Figure 1.

Article	Problem type	Resolution procedure	Orientation	Weight	Load Balance	Stability	Time-Dependent
(W. Zhu and Lim 2012)	SLOP-SKP	TSM	X	X			
(Gonçalves and Resende 2013)	SBSBPP	BRKGA	X	X		X	
(Paquay, Schyns, and Limbourg 2016)	SBSBPP	EM	X	X	X	X	
(Mahvash, Awasthi, and Chauhan 2018)	SBSBPP	CG	X	X			
(Alonso, Alvarez-Valdes, and Parreño 2020)	SSSCSP	GRASP	X	X	X	X	X
(Tresca et al. 2022)	SBSBPP	MH	X	X	X	X	
(Harrath 2022)	SBSBPP	H	X	X	X		
(Li, M. Chen, and Huo 2022)	SBSBPP-MSSCSP	HALNs	X	X		X	
(M. Chen, Huo, and Duan 2023)	BSDPP	HBBO	X	X			
3D-SCPTOP	SSSCSP & MILOPP	MH	X	X	X	X	X

Abbreviations. **TSM** (Tree search method), **BRKGA** Biased Randomized Jey Genetic Algorithm; **EM** (Exact Methods), **MH** (MatHeuristic), **CG** (Column Generation), **H** (Heuristic); **HCLP** (Hybrid Adaptive Large Neighborhood Search), **HBBO** (Hybrid Biogeography-Based Optimization)

Table 3: Comparison of C&P problems

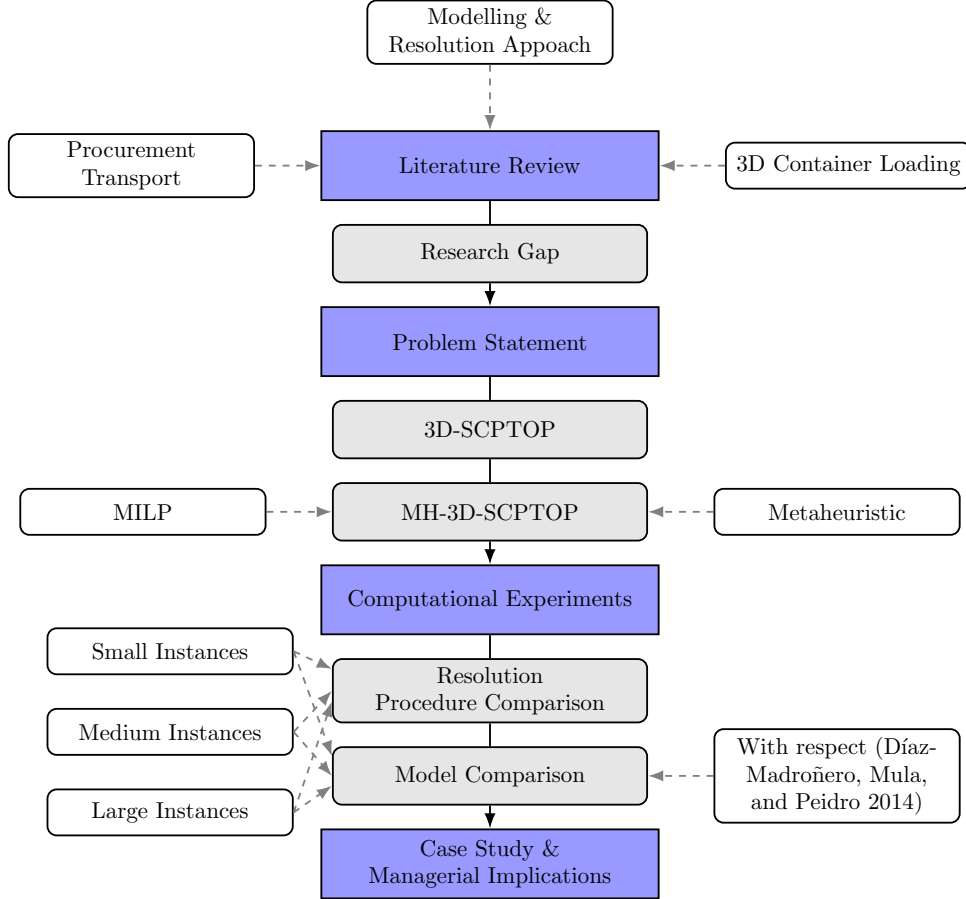


Figure 1: Research Methodology

3 Problem Description

The objective is to minimise the transportation and inventory holding costs of a manufacturing site by integrating procurement transport, RTI management and inventory decisions with full-truckload (FTL) loading constraints expressed in a stack-based form. The supply chain under study has a single source and a single destination. Products are shipped in different RTI types. RTIs are not placed individually in three dimensions but are grouped into vertical *stacks* whose 2D footprints are placed on the truck floor. Each stack type has a fixed footprint and is compatible with a subset of RTI types; each layer of a stack is homogeneous (all RTIs in a layer share a single type), while different layers of the same stack may carry different compatible RTI types. The total stack height is bounded by the truck height and the weight borne by the bottom of each stack is bounded by a crush-limit cap. Trucks carry full RTIs from the supplier to

the plant and return folded empties from the plant to the supplier; folding affects RTI height, and hence the number of layers that fit in a stack, but not the stack footprint.

The assembler's demand drives the procurement process because it determines the optimal quantity of each product to procure from the supplier to the manufacturing site. Raw materials and assembled parts are produced at the supplier, packed in their corresponding RTI, and transported under FTL. Each RTI carries a single product type at any given time, and each product is shipped in a single RTI type. Once full RTIs arrive at the plant, the products are consumed and the RTIs are emptied, folded and returned to the supplier.

Stacks must not overlap and must be contained within the truck's floor; the truck's total weight is bounded; and the per-stack weight cap is respected with *period-exact* knowledge of the products loaded in each slot, so that crush-limit tests reflect the actual period's load rather than a worst-case bound. Stacks may be rotated by 90° on the truck floor. Because stacks are self-supporting by construction, no additional vertical-stability constraints are required. As FTL is the mode of transport, trucks must reach a minimum volume utilisation measured on the RTI volumes effectively loaded. Minimum coverage stock is enforced on future demand.

Initial data.

- **Product data:** initial inventory, periodic demand (in full RTIs) at the plant and at the supplier, product-to-RTI mapping, weight of each full RTI.
- **RTI data:** height when full and when folded, weight of a folded RTI, volume.
- **Stack data:** footprint (p_h, q_h) of each stack type $h \in H$, set of compatible RTI types $R(h)$, and, for each $r \in R(h)$, number of RTIs per layer n_{hr} .
- **Transport data:** number of available trucks, truck length/width/height, truck weight capacity, per-stack weight cap, minimum truck volume utilisation.

To determine.

- A set of loading configurations (LCs), each specifying the stack-type assignment, 2D placement and orientation of its slots, and the number of layers of each compatible RTI type in each slot.
- The number of trucks used per day and the LC they employ.
- The number of full RTIs of each product loaded in each slot per period (plant-bound trucks) and the number of folded RTIs of each type loaded per truck (supplier-bound trucks).
- The resulting inventory of full and folded RTIs at both nodes.

Subject to.

- 2D non-overlap and containment of stacks on the truck floor, with allowed 90° rotation.
- Per-stack height budget.
- Per-stack crush-limit weight cap, period-exact on plant-bound trucks and deterministic on supplier-bound trucks.
- Per-truck weight capacity.
- Minimum truck volume utilisation.
- Minimum coverage stock.

In order to:

- Minimise total transportation cost.
- Minimise total inventory holding cost.

4 Mathematical Model

The 3D-SCPTOP formulation integrates stack-based truck-loading constraints with procurement transport and inventory decisions. It departs from a standard 3D container-loading formulation in two related ways. First, the 3D placement of individual RTIs inside the truck is replaced by a 2D placement of variable-composition *stacks* on the truck floor, with per-stack height and weight caps. Second, the per-period

decision of which product fills each slot is retained, but the crush-limit cap is evaluated slot-by-slot using the period's actual product mix. This yields a tighter per-stack weight bound than a worst-case-per-RTI-type abstraction, while eliminating vertical-stability constraints altogether, since stacks are self-supporting by construction.

Notational conventions. Throughout this section, $g \in G = \{s, p\}$ indexes nodes (supplier and plant), $n \in N$ products, $r \in R$ RTI types, $h \in H$ stack types, $i, j \in I$ stack-slot positions within a loading configuration, $l \in L$ loading configurations, $k \in K$ trucks and $t \in T$ planning periods. Variables whose direction is determined by definition omit the g subscript: V_{klnt} and $V_{wslot_{klnt}}$ denote full-RTI loads on plant-bound trucks, and V_{klrt}^e denotes folded-RTI loads on supplier-bound trucks. Table 4 collects the complete nomenclature.

Table 4: Nomenclature of 3D-SCPTOP (stack-based reformulation)

Indices and sets	
G	Supplier and plant nodes, $G = \{s, p\}$.
N	Products, $n = 1, \dots, N$.
R	RTI types, $r = 1, \dots, R$.
H	Stack types, $h = 1, \dots, H$.
I	Stack-slot indices within a loading configuration, $i = 1, \dots, I $ (see eq. 4.1).
L	Loading configurations, $l = 1, \dots, L$.
K	Trucks, $k = 1, \dots, K$.
T	Planning periods, $t = 1, \dots, T$.
$M(r)$	Products carried by RTI type r , $M(r) \subseteq N$.
$R(h)$	RTI types compatible with stack type h , $R(h) \subseteq R$.
Parameters	
cK	Cost of using a truck.
lt	Lead transport time between plant and supplier (and vice versa).
i_{gn}	Initial inventory of $n \in N$ at $g \in G$ (full RTIs).
ie_{gr}	Initial inventory of folded RTIs $r \in R$ at $g \in G$.
ic_{gn}	Inventory cost of $n \in N$ at $g \in G$.
d_{gnt}	Production plan for $n \in N$ at $g \in G$ in $t \in T$.
q_{nt}	Initial arrivals of full RTIs of product n at p in $t \in \{1, \dots, lt\}$.
q_{rt}	Initial arrivals of folded RTIs of type r at s in $t \in \{1, \dots, lt\}$.
ss_{gnt}	Security stock of n at g in t (full RTIs).
ss_{grt}	Security stock of folded RTIs r at g in t .
w_n	Weight of a full RTI carrying product n .
w_r^f	Weight of a folded RTI of type r .
pK^w	Maximum truck weight capacity.
$W^{\max}$	Per-stack crush-limit weight cap.
pK, qK, rK	Truck length, width, and height.
Γ_K	Truck volume, $\Gamma_K = pK \cdot qK \cdot rK$.
α	Minimum truck volume utilisation ($0 < \alpha \leq 1$).
p_h, q_h	Footprint length and width of stack type h .
r_{gr}	Height of one RTI of type r at node g (full at p , folded at s).
γ_{gr}	Volume of one RTI of type r at node g .
n_{hr}	RTIs of type $r \in R(h)$ per layer of stack type h .
M	A sufficiently large constant.
Decision variables – LC level	
Z_{gl}	$\in \{0, 1\}$. Equals 1 iff LC l is active in direction g .
X_{gihl}	$\in \{0, 1\}$. Equals 1 iff slot i of LC l is of stack type h in direction g .
L_{gihlr}	$\in \mathbb{Z}_{\geq 0}$. Number of layers of RTI type $r \in R(h)$ in slot i of LC l in direction g .
LX_{gihl}	$\in \{0, 1\}$. Equals 1 iff stack h 's length is along the x -axis in slot i .
LY_{gihl}	$\in \{0, 1\}$. Equals 1 iff stack h 's length is along the y -axis in slot i .
PX_{gil}, PY_{gil}	≥ 0 . Bottom-left coordinates of slot i in LC l in direction g .
A_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is to the left of slot j in LC l .
B_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is to the right of slot j in LC l .
C_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is behind slot j in LC l .

D_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is in front of slot j in LC l .
Decision variables – runtime level	
Y_{gkt}	$\in \{0, 1\}$. Equals 1 iff truck k is used in direction g in period t .
Y_{gklt}	$\in \{0, 1\}$. Equals 1 iff truck k uses LC l in direction g in period t .
$V_{kln t}$	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of product n on plant-bound truck k using LC l in period t .
V_{kilnt}^{slot}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of product n in slot i of plant-bound truck k using LC l in period t .
V_{klrt}^e	$\in \mathbb{Z}_{\geq 0}$. Folded RTIs of type r on supplier-bound truck k using LC l in period t .
I_{gnt}	$\in \mathbb{Z}_{\geq 0}$. End-of-period inventory of n at g in t (full RTIs).
Ie_{gtr}	$\in \mathbb{Z}_{\geq 0}$. End-of-period inventory of folded RTIs r at g in t .
Q_{nt}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of product n in transit to the plant in t .
Q_{rt}	$\in \mathbb{Z}_{\geq 0}$. Folded RTIs of type r in transit to the supplier in t .

Objective

$$\text{Min } z = \sum_{g \in G} \sum_{k \in K} \sum_{t \in T} cK \cdot Y_{gkt} + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} ic_{gn} \cdot I_{gnt} \quad (1)$$

Inventory and flow balance

The inventory balance block is structurally identical to the original formulation. For completeness it is reproduced below.

$$I_{pnt} = I_{p,n,t-lt} - d_{pnt} + Q_{n,t-lt} \quad \forall n \in N, t \geq lt + 1 \quad (2)$$

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + q_{nt} \quad \forall n \in N, t \in \{2, \dots, lt\} \quad (3)$$

$$I_{pnt} = i_{pn} - d_{pnt} + q_{nt} \quad \forall n \in N, t = 1 \quad (4)$$

$$Ie_{ptr} = Ie_{p,t-1,r} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \quad \forall r \in R, t \geq 2 \quad (5)$$

$$Ie_{ptr} = ie_{pr} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \quad \forall r \in R, t = 1 \quad (6)$$

$$I_{snt} = I_{s,n,t-1} + d_{snt} - Q_{nt} \quad \forall n \in N, t \geq 2 \quad (7)$$

$$I_{snt} = i_{sn} + d_{snt} - Q_{nt} \quad \forall n \in N, t = 1 \quad (8)$$

$$Ie_{str} = Ie_{s,t-lt,r} + Q_{r,t-lt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t \geq lt + 1 \quad (9)$$

$$Ie_{str} = Ie_{s,t-1,r} + q_{rt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t \in \{2, \dots, lt\} \quad (10)$$

$$Ie_{str} = ie_{sr} + q_{rt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t = 1 \quad (11)$$

$$I_{gnt} \geq ss_{gnt}, \quad Ie_{gtr} \geq ss_{gtr} \quad \forall g \in G, n \in N, r \in R, t \in T \quad (12)$$

Stack packing constraints

Exactly one stack type can be assigned to a slot, and only if its LC is active:

$$\sum_{h \in H} X_{ghl} \leq Z_{gl} \quad \forall g \in G, i \in I, l \in L. \quad (13)$$

The stack occupying slot i is oriented either with its length along the x -axis or along the y -axis:

$$LX_{ghl} + LY_{ghl} = X_{ghl} \quad \forall g \in G, h \in H, i \in I, l \in L. \quad (14)$$

The total stack height cannot exceed the truck height. The following constraint also forces $L_{ghlr} = 0$ whenever $X_{ghl} = 0$, since $r_{gr} > 0$:

$$\sum_{r \in R(h)} r_{gr} L_{ghlr} \leq rK \cdot X_{ghl} \quad \forall g \in G, h \in H, i \in I, l \in L. \quad (15)$$

Slots must not overlap. For each pair $i \neq j$ in LC l :

$$PX_{gil} + \sum_{h \in H} (p_h LX_{ghl} + q_h LY_{ghl}) \leq PX_{gjl} + M(1 - A_{gijl}) \quad (16)$$

$$PX_{gjl} + \sum_{h \in H} (p_h LX_{ghl} + q_h LY_{ghl}) \leq PX_{gil} + M(1 - B_{gijl}) \quad (17)$$

$$PY_{gil} + \sum_{h \in H} (q_h LX_{ghl} + p_h LY_{ghl}) \leq PY_{gjl} + M(1 - C_{gijl}) \quad (18)$$

$$PY_{gjl} + \sum_{h \in H} (q_h LX_{ghl} + p_h LY_{ghl}) \leq PY_{gil} + M(1 - D_{gijl}) \quad (19)$$

with the cover constraint ensuring that whenever both slots are used at least one relative-position variable is active:

$$A_{gijl} + B_{gijl} + C_{gijl} + D_{gijl} \geq \sum_h X_{ghl} + \sum_h X_{ghl} - 1 \quad \forall g, l, i \neq j. \quad (20)$$

Containment in the truck floor:

$$PX_{gil} + \sum_{h \in H} (p_h LX_{ghl} + q_h LY_{ghl}) \leq pK + M(1 - \sum_h X_{ghl}) \quad \forall g, i, l, \quad (21)$$

$$PY_{gil} + \sum_{h \in H} (q_h LX_{ghl} + p_h LY_{ghl}) \leq qK + M(1 - \sum_h X_{ghl}) \quad \forall g, i, l. \quad (22)$$

Minimum truck volume utilisation, measured on the actual RTI volumes (kept in RTI-volume form so non-full layers or fillers can be introduced in later extensions):

$$\sum_{i \in I} \sum_{h \in H} \sum_{r \in R(h)} \gamma_{gr} n_{hr} L_{ghlr} \geq \alpha \Gamma_K Z_{gl} \quad \forall g \in G, l \in L. \quad (23)$$

Truck-LC linking and runtime loading

A truck uses at most one LC in a period, and can only use an active LC:

$$\sum_{l \in L} Y_{gklt} = Y_{gkt} \quad \forall g, k, t \quad (24)$$

$$Y_{gklt} \leq Z_{gl} \quad \forall g, k, l, t \quad (25)$$

For plant-bound trucks, full RTIs of type r loaded in slot i cannot exceed the layer capacity of the stack placed in that slot; products sharing the same RTI type partition that slot-capacity:

$$\sum_{n \in M(r)} V_{kilnt}^{\text{slot}} \leq \sum_{h: r \in R(h)} n_{hr} L_{p,i,h,l,r} + M(1 - Y_{pkl}) \quad \forall k, i, l, r, t. \quad (26)$$

Truck-level aggregation of per-slot loads:

$$V_{klnt} = \sum_{i \in I} V_{kilnt}^{\text{slot}} \quad \forall k, l, n, t. \quad (27)$$

For supplier-bound trucks, folded RTIs of type r are capped by the aggregate layer capacity of the LC in use (no slot-level tracking is needed because folded-RTI weight is deterministic in r):

$$V_{klrt}^e \leq \sum_{i \in I} \sum_{h: r \in R(h)} n_{hr} L_{s,i,h,l,r} + M(1 - Y_{skt}) \quad \forall k, l, r, t. \quad (28)$$

Weight constraints

Per-truck weight, plant-bound:

$$\sum_{l \in L} \sum_{n \in N} w_n V_{klnt} \leq pK^w Y_{pkt} \quad \forall k, t. \quad (29)$$

Per-truck weight, supplier-bound:

$$\sum_{l \in L} \sum_{r \in R} w_r^f V_{klrt}^e \leq pK^w Y_{skt} \quad \forall k, t. \quad (30)$$

Per-stack crush-limit, period-exact on plant-bound trucks (the cap reflects the actual product mix loaded in each slot):

$$\sum_{n \in N} w_n V_{kilnt}^{\text{slot}} \leq W^{\max} \quad \forall k, i, l, t. \quad (31)$$

Per-stack crush-limit on supplier-bound trucks (deterministic in the LC since folded-RTI weight depends only on r):

$$\sum_{h \in H} \sum_{r \in R(h)} w_r^f n_{hr} L_{s,i,h,l,r} \leq W^{\max} \quad \forall i, l. \quad (32)$$

Flow aggregation

$$Q_{nt} = \sum_{k \in K} \sum_{l \in L} V_{klnt} \quad \forall n, t \quad (33)$$

$$Q_{rt} = \sum_{k \in K} \sum_{l \in L} V_{klrt}^e \quad \forall r, t \quad (34)$$

4.1 Symmetry breaking and slot-set sizing

The slots $i \in I$ of a given LC are interchangeable, so any solution admits $|I|!$ permutations of the slot index. We break this symmetry by enforcing that slot $i + 1$ is used only if slot i is used:

$$\sum_{h \in H} X_{g,i+1,h,l} \leq \sum_{h \in H} X_{g,i,h,l} \quad \forall g \in G, l \in L, i \in \{1, \dots, |I| - 1\}. \quad (35)$$

The cardinality of I is a modelling parameter. An upper bound that eliminates provably unused slot variables without cutting any feasible layout is

$$|I| \leq \left\lfloor \frac{pK \cdot qK}{\min_{h \in H} p_h \cdot q_h} \right\rfloor, \quad (36)$$

corresponding to tiling the truck floor with the smallest-footprint stack type; it serves as a tight preprocessing limit on $|I|$.

Domains

$$\begin{aligned} Z_{gl}, X_{gihl}, LX_{gihl}, LY_{gihl}, A_{gijl}, B_{gijl}, C_{gijl}, D_{gijl}, Y_{gkt}, Y_{gklt} &\in \{0, 1\}, \\ L_{gihlr}, V_{klnt}, V_{kilnt}^{\text{slot}}, V_{klrt}^e, I_{gnt}, I_{gtr}, Q_{nt}, Q_{rt} &\in \mathbb{Z}_{\geq 0}, \\ PX_{gil}, PY_{gil} &\geq 0. \end{aligned}$$

5 Matheuristic Procedure

Because the slot-placement subproblem is a constrained 2D cutting-and-packing problem and the per-slot layer-composition subproblem is a small integer knapsack, an exact solve of the full model is intractable for industrial instances. We retain the decomposition structure of the original matheuristic – enumerate or construct a pool of loading configurations, then let a reduced MILP pick configurations, assign trucks and products, and manage inventory – but replace the 3D packing subroutine by a two-step stack-based scheme.

5.1 PLC-SCPTOP

PLC-SCPTOP operates on a precomputed pool of LCs. Each LC l in direction g exposes its aggregate RTI-type capacity through the parameter

$$U_{glr} = \sum_{i \in I} \sum_{h : r \in R(h)} n_{hr} L_{gihlr}^*, \quad \forall g \in G, l \in L, r \in R, \quad (37)$$

where L_{gihlr}^* are the fixed layer counts of the generated LC. This parameter takes the place of the former $lc_{lr,g}$ (“number of containers of type c in configuration l ”). Apart from this substitution, the structure of

PLC-SCPTOP is unchanged: the inventory-balance, safety-stock, one-LC-per-truck, and flow-aggregation constraints carry over verbatim, with the capacity-linking constraints rewritten to use U_{glr} .

Because PLC-SCPTOP does not carry slot-level information, it enforces the per-stack crush cap only in a conservative form using the worst-case product weight per RTI type, $\bar{w}_r = \max_{n \in M(r)} w_n$. The period-exact per-stack cap (31) is enforced at the LC-generation stage, where the pool is restricted to layer compositions feasible for every product mix the plant might load.

5.2 Stack-column generation and 2D placement

The LC generator decomposes the packing subroutine into two cleanly separated steps.

Step 1: stack-column generation. For each stack type $h \in H$ and each direction $g \in G$, enumerate or column-generate all layer vectors $\lambda_h = (\lambda_r)_{r \in R(h)} \in \mathbb{Z}_{\geq 0}^{|R(h)|}$ satisfying

$$\begin{aligned} \sum_{r \in R(h)} r_{gr} \lambda_r &\leq rK, \\ \sum_{r \in R(h)} \bar{w}_r n_{hr} \lambda_r &\leq W^{\max} && \text{(plant direction, conservative bound),} \\ \sum_{r \in R(h)} w_r^f n_{hr} \lambda_r &\leq W^{\max} && \text{(supplier direction, deterministic).} \end{aligned}$$

Each feasible λ_h is a *stack column*: a candidate internal composition of a slot of type h .

Step 2: 2D slot placement. Given a target mix of stack columns (quantities per stack type, weighted by current inventory deficits as in Algorithms 1–2), place the selected columns on the truck floor by solving a 2D-SLOPP with 90° rotation over the footprints (p_h, q_h) . Any standard 2D packing heuristic (constructive, GRASP, BRKGA) or MILP formulation may be used as the inner solver.

Algorithm structure. The outer matheuristic is unchanged but calls the two-step generator above in place of the former 3D packing routine.

Algorithm 1 Initial Loading Configurations (stack-based)

```

 $T$                                 ▷ list of time periods
 $\Delta$                                 ▷ list of look-ahead horizons
LC  $\leftarrow \emptyset$ 
for  $t \in T$  do
  for  $\delta \in \Delta$  do
    IncomingTransport( $t$ )
    Produce( $t$ )
    while TransportResourceAvailable do
       $\mathcal{C} \leftarrow \text{StackColumnsNeeded}(t, \delta, \text{LC})$           ▷ Step 1: required per-type layer vectors
       $P \leftarrow \text{Place2D}(\mathcal{C})$                                 ▷ Step 2: 2D-SLOPP with rotation
      LC.append( $P$ )
    end while
  end for
end for
LC  $\leftarrow \text{EliminateDuplicatesAndDominated}(\text{LC})$ 
return LC

```

Here the 3D-SCPTOP model is validated and compared to the work by (Díaz-Madroño, Mula, and Peidro 2014). This fuzzy optimisation model integrates procurement transport and inventory holding decisions by contemplating transport capacity restrictions by a maximum number of containers per truck, which we refer to as SCPTOP from this point onwards. To compare the SCPTOP model, we make three key modifications: fuzzy parameters are not considered; transport capacity restriction takes into account the volume of individual containers and the truck's capacity; products can be loaded in more than one container type. The SCPTOP model is extended and defined in Appendix A. We evaluate and compare both the

Algorithm 2 Refinement Loading Configurations (stack-based)

```
TimePeriods, TimeSpan ▷ user-defined
for  $t \in \text{TimePeriods}$  do
  for  $\delta \in \text{TimeSpan}$  do
     $\mathbf{I} \leftarrow \text{SolvePLCSCPTOP}(\text{LC}, t)$ 
     $\mathcal{C} \leftarrow \text{StackColumnsNeeded}(\mathbf{I}, t, \delta, \text{LC})$ 
     $P \leftarrow \text{Place2D}(\mathcal{C})$ 
     $\text{LC.append}(P)$ 
  end for
end for
 $\text{LC} \leftarrow \text{EliminateDuplicatesAndDominated}(\text{LC})$ 
return LC
```

Algorithm 3 MATH-3D-SCPTOP (stack-based)

```
1:  $\Delta_1, \Delta_2$  ▷ look-ahead horizons
2:  $\text{LC} \leftarrow \text{InitialLoadingConfigurations}(\Delta_1)$ 
3:  $\text{LC} \leftarrow \text{RefinementLoadingConfigurations}(\text{LC}, \Delta_2)$ 
4: Build  $U_{glr}$  from LC via (37)
5:  $\text{SolvePLCSCPTOP}(\text{LC}, |T|)$ 
```

3D-SCPTOP and SCPTOP models, in terms of solution quality, transport costs, inventory costs and computational times. The 3D-SCPTOP and SCPTOP mathematical formulations are solved for all the instance sets with CP-SAT from OR-Tools and Gurobi using their default settings. Additionally, as $(C\&P)$ problems are NP-hard, the Meta-3D-SCPTOP algorithm has been developed to find igh-quality solutions for large real-world instances, based on a case study. So the performance of the 3D-SCPTOP, the Meta-3D-SCPTOP and the SCPTOP models is compared and evaluated as follows.

Twenty-four instances are artificially generated by a stochastic process, whose key characteristics are given as follows:

- **Demand Generation:** the demand for each product is randomly generated by deviating from a given mean by plus or minus ten percent. This introduces variability into the demand values for different products
- **Time Periods:** Half of the instances have a total of five time periods, while the other half have a total of ten time periods. This allows evaluating the model's performance over different time horizons.
- **Container Types:** Each instance includes a varying number container types, specifically two, three or four. The distribution of container types is the same among instances, which implies that an equal number of instances has the same number of containers
- **Container Dimensions:** The dimensions of each container type are randomly generated. The width, length and height of boxes are fractions of the width, length, or height of trucks. This randomness ensures diversity in the sizes of containers across the generated instances. An additional constraint is applied to the generation of containers. The volume of each container must be at least $1/n$ of the truck's volume, where n can take the values of four, six or eight. This ensures that containers are appropriately sized in relation to trucks

By incorporating these stochastic elements into the generation process, the 24 instances provide a diverse set of scenarios for evaluating the model's performance. Variability in demand, time periods, box types and box dimensions introduces realistic and challenging conditions that the model needs to address. Furthermore, we use the number of time periods, the number of box types, the number of products and the maximum amount of containers needed to load a truck (in volume terms) as explanatory variables of various metrics from the models, such as total cost, transport cost, inventory cost, solution gap and solution time. The instance name is in the form of $L_n.t.c.v$, where n is product type, t is the number of time periods, c is the number of container types and v is the minimum volume of containers.

Each test is limited to 4 hours (14,400 seconds) of run-time per instance. Tests are performed on a Microsoft Windows Server 2022 Standard n with an AMD EPYC 7402 24-Core Processor running at 2,800 MHz, with 256 GB of RAM using Python as the programming language.

6 Practical application

This section presents a practical procurement transport application. The proposed resolution framework was evaluated using real data from an automotive industry supply chain. We validate and compare the 3D-SCPTOP solution procedures to the company’s procedure, which employs predefined loading configurations that indicate the number of each container type to be carried. Finding these loading configurations is a very tedious and time-consuming manual task that is based on a spreadsheet procedure. A mathematical programming model, dubbed as PLC-SCPTOP, which replicates the current’s company procedure, is then utilised to minimise the total cost, with a summation of procurement transport and inventory holding costs, and one based on these predetermined configurations.

The real-world data derived from the company’s operations act as the empirical foundation for examining both the PLC-SCPTOP and the Meta-3D-SCPTOP. To broaden the scope of this analysis, we scale and manipulate these data, guided by a stochastic process, to simulate several potential operational scenarios. Setting our analysis to mirror the dynamic and unpredictable realities of supply chain operations thus enhances the validity of our evaluation and the practical relevance of the suggested metaheuristic. For this purpose, we propose three different scenarios for the created instances: normal demand, low variability demand, high variability demand. Table ??, found in the Appendix A outlines the models’ performance across these instances. Table 5 summarises models’ performance and resolution times.

Table 5: Summary of performance for the practical application

Model	Objective Value	Best Bound	Duration (seconds)
MathH-3D-SCTOP	252235.198934	247307.820849	740.000000
Meta-3D-SCTOP	256715.145777	NaN	677.555556
PLC-SCTOP	275226.680415	271798.477640	44.670370

Figures ?? - ?? describe the performance of both the models and solution procedures in the different scenarios. The x-axis shows the scenario, while the y-axis depicts the objective value, inventory costs, or transport costs. Each colour represents a different model applied to obtain the solution.

The results reveal that an effective systematic method is proposed for solving procurement transport operational planning with 3D full-truck loading constraints. This solution is both automated and executed within a reasonable computational timeframe. Our methodology proves particularly beneficial when there is significant discrepancy between the company’s proposed loading configurations and the actual demand concerning container types. In such instances, the company is usually left with two choices: either use the fixed loading configurations and, thus, incur higher inventory and transport costs, or manually resolve the loading configuration via a spreadsheet procedure, which might lead to errors. However, one potential disadvantage of the proposed Meta-3D-SCPTOP is its performance under normal demand conditions because the demand for different container types is consistent with the loading configurations, but the metaheuristic may result in fewer optimal inventory holding costs compared to the company’s current procedure. This is because the Meta-3D-SCPTOP obtains decisions locally, while exact methods are guaranteed, given enough time, to find the optimal solution.

This issue is, however, solved when utilising MathH-3D-SCTOP because it effectively combines both heuristic and exact methods. While the heuristic provides a fast and reasonably good solution for the initial loading configurations, the exact method improves it by further lowering inventory and transport costs.

Therefore, this method shows promising results for reducing inventory and transport costs, particularly under variable demand and misalignment between loading configurations and container types.

7 Conclusions

We herein propose a MILP model, 3D-SCPTOP, to tackle procurement transportation planning at an operational level by incorporating 3D loading constraints into its considerations. Given the NP-hard nature of 3D bin packing, we have also developed a metaheuristic approach, Meta-3D-SCPTOP, which is later refined and enhanced by employing a mathheuristic technique, MatH-3D-SCPTOP.

A case study has also been conducted within an automotive SC that comprises a car assembler, a first-tier supplier and a second-tier supplier. The objective of the 3D-SCPTOP model is twofold: (i) to minimise the total number of trucks used; (ii) to minimise inventory holding costs. This approach takes into account the

complexities of supply chain logistics in the automotive industry by demonstrating a successful integration of procurement transport planning and 3D loading constraints.

Managerial implications in a SC context are oriented to the proposed solution methodology, improving the solutions found by the company's procedure. It is also able to adapt to demand changes while finding high-quality solutions in reasonable computational times.

Nevertheless, several challenges remain for further research lines to deal with; for instance, the inclusion of multiple suppliers to better represent complex supply chains. A more comprehensive transport operation model could be achieved by incorporating routing aspects, such as vehicle routing with 3D loading constraints. Furthermore, the introduction of dynamic stability into loading constraints may also yield more practical, robust solutions. Integrating sourcing and delivery transport decisions could improve supply chain efficiency. Finally, the incorporation of production scheduling into the model has the potential of offering a cohesive approach to supply chain planning optimisation by aligning production timelines with transportation and inventory strategies.

A Mathematical Model

Table 6: Nomenclature of Supply Chain Operation Transport Planning

Sets of indices	
$I :$	Set of products ($i = 1, 2, \dots, I$)
$J :$	Set of groups composed of products that must be ordered together ($j = 1, 2, \dots, J$)
$K :$	Set of trucks ($k = 1, 2, \dots, K$)
$T :$	Set of planning periods (days) ($t = 1, 2, \dots, T$)
Decision variables	
$Q_{ikt} :$	Units transported of i by k in period t (units)
$G_{ijkt} :$	Units transported of i corresponding to group j by k in period t (units)
$C_{kt} :$	Volume of containers transported by k in period t (units)
$I_{it} :$	Inventory amount of i at the end of period t (units)
$K_{jkt} :$	Number of lots to order of products of group j by k in period t
$Y_{kt} :$	Binary variable indicating whether a truck k has been used in period t
Objective functions	
$z_1 :$	Total number of trucks utilized
$z_2 :$	Total inventory amount generated
Parameters	
$u_i :$	Amount of product i that fits in a container (units)
$l_j :$	Number of units of each group lot j (units)
$b_{ij} :$	1 if product i belongs to group j , and 0 otherwise
$D_{it} :$	Demand of product i in t (units) (considered firm)
$\Gamma_k :$	Volume of truck k
$\Gamma_i :$	Volume of container carrying product i
$m :$	Minimum truck occupation (in containers)
$I0_i :$	Inventory amount of i in period 0
$IC_i :$	Inventory cost of i
$C_k :$	Cost of using truck k
$SC_i :$	Number of days of stock coverage for product i

Minimize the total cost transport and inventory

$$\text{Min } z \cong \sum_{k=1}^K \sum_{t=1}^T Y_{kt} + \sum_{i=1}^I \sum_{t=1}^T I_{it} \cdot IC_i$$

subject to

$$\begin{aligned} I_{it} &= I_{i(t-1)} - D_{it} + \sum_{k=1}^K Q_{ikt} \quad \forall i \in I, t \in T, \\ Q_{ikt} &= \sum_{j=1}^J G_{ijkt} \quad \forall i \in I, k \in K, t \in T, \\ G_{ijkt} &= K_{jkt} \cdot l_j \cdot b_{ij} \quad \forall i \in I, j \in I, k \in K, t \in T, \\ C_{kt} &= \sum_{i=1}^I Q_{ikt} / u_i \cdot \Gamma_i \quad \forall k \in K, t \in T, \\ C_{kt} &\leq \Gamma_k \cdot Y_{kt} \quad \forall k \in K, t \in T, \\ C_{kt} &\geq m \cdot \Gamma_k Y_{kt} \quad \forall k \in K, t \in T, \\ I_{nt} &\geq \begin{cases} \sum_{s \in \{1, \dots, SC_i\}} D_{i,t+s} & \text{if } i \in I, t \in \{1, \dots, |T| - SC_i\} \\ \sum_{s \in \{1, \dots, SC_i\}} D_{i,|T|-SC_i+s} & \text{other wise} \end{cases} \\ Y_{kt} &\leq 1 \quad \forall k, t, \\ I_{it}, Q_{ikt}, G_{ijkt}, C_{kt}, K_{jkt}, Y_{kt} &\geq 0 \text{ integer} \quad \forall i \in I, j \in I, k \in K, t \in T. \end{aligned}$$

In our case $|J| = |I|$, one group for each product.

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